

Divide and Conquer – Searching & Basic Problems

Binary Search, Finding Maximum and Minimum using Divide and Conquer Dr.

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Question 1: Search an Element in a Sorted Array

Given a sorted array of integers and a key value, use the Binary Search (Divide and Conquer) method to determine whether the key is present in the array. If found, print its position; otherwise, print "Element not found".

Test Case 1

Input

5

10 20 30 40 50

30

Expected Output

Element found at position 3

Test Case 2

Input

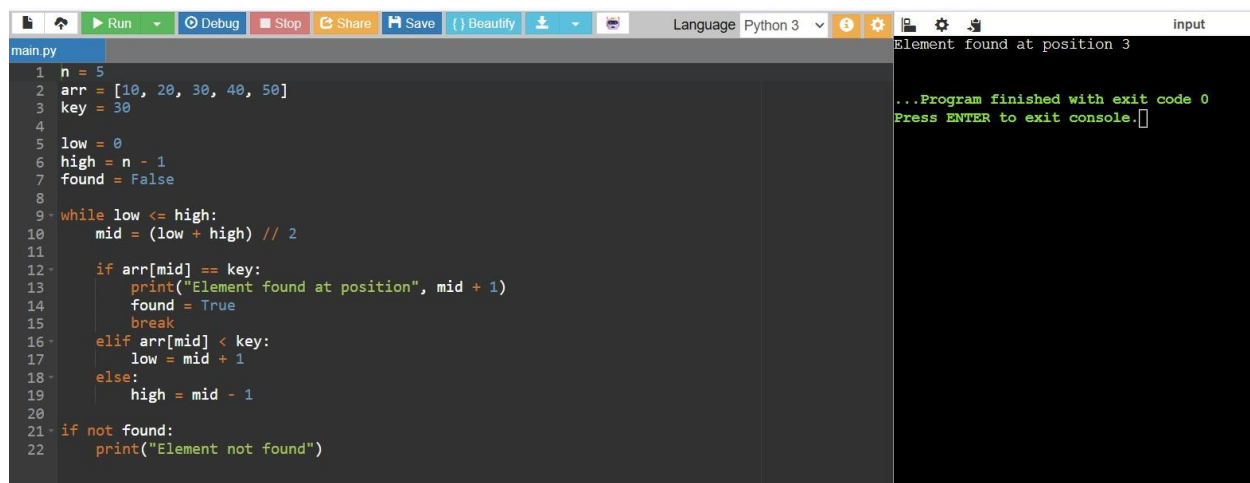
5

10 20 30 40 50

25

Expected Output

Element not found



The screenshot shows a Python IDE with a file named 'main.py'. The code implements a binary search algorithm on a sorted array [10, 20, 30, 40, 50] with a key value of 30. The algorithm uses a while loop to find the element, and if found, it prints the position (index + 1). The output in the console shows 'Element found at position 3' and a message indicating the program finished with exit code 0.

```
1 n = 5
2 arr = [10, 20, 30, 40, 50]
3 key = 30
4
5 low = 0
6 high = n - 1
7 found = False
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         print("Element found at position", mid + 1)
14         found = True
15         break
16     elif arr[mid] < key:
17         low = mid + 1
18     else:
19         high = mid - 1
20
21 if not found:
22     print("Element not found")
```

Element found at position 3

...Program finished with exit code 0
Press ENTER to exit console.

Question 2: Find the Index of a Given Number

Given a sorted array and a target number, find the index of the target using Binary Search. Assume array indexing starts from 0.

Test Case 1

Input

6
2 4 6 8 10 12
8

Expected Output

Index = 3

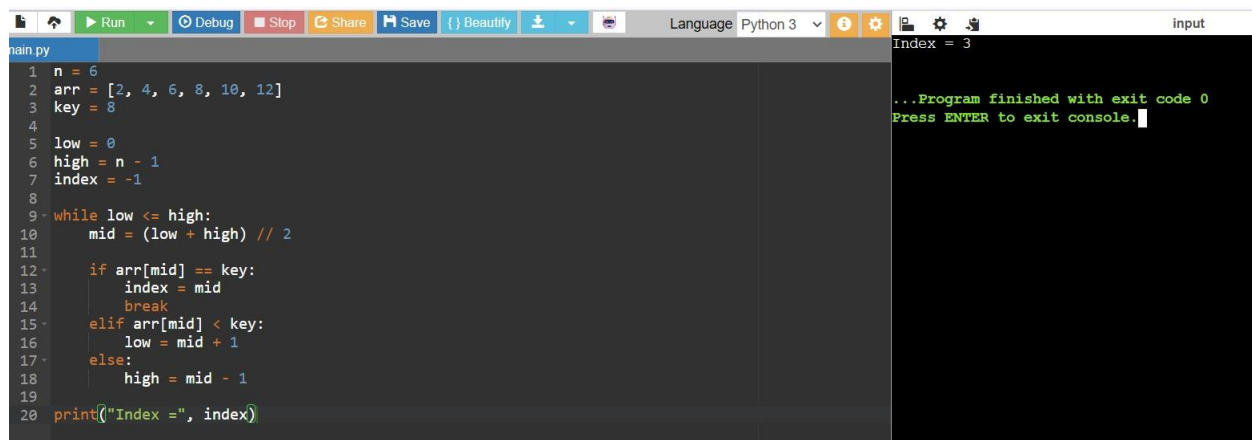
Test Case 2

Input

6
2 4 6 8 10 12
5

Expected Output

Index = -1



The screenshot shows a Python IDE with a file named 'main.py'. The code implements a binary search algorithm. It initializes 'n' to 6, 'arr' to [2, 4, 6, 8, 10, 12], and 'key' to 8. It then uses a while loop to find the index of the key. The output in the console is 'Index = 3'. Below the console output, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.'.

```
1 n = 6
2 arr = [2, 4, 6, 8, 10, 12]
3 key = 8
4
5 low = 0
6 high = n - 1
7 index = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         index = mid
14         break
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 print("Index =", index)
```

Index = 3

...Program finished with exit code 0
Press ENTER to exit console.

Question 3: Count the Number of Iterations in Binary Search

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Test Case 1

Input

7
5 10 15 20 25 30 35
25

Expected Output

Element found
Iterations = 2

Test Case 2

Input

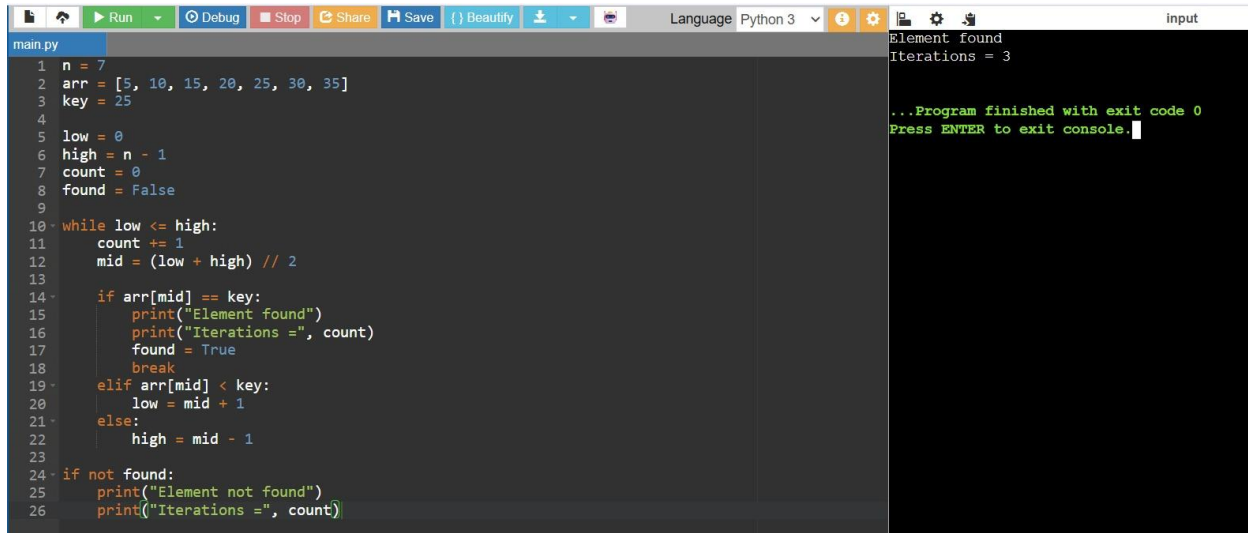
7
5 10 15 20 25 30 35

40

Expected Output

Element not found

Iterations = 3



The screenshot shows a Python IDE with a file named 'main.py'. The code implements a binary search algorithm on a sorted array [5, 10, 15, 20, 25, 30, 35] with a target key of 25. The algorithm uses a while loop to find the element, and if not found, it prints 'Element not found' and the number of iterations. The output in the console shows 'Element found' and 'Iterations = 3', followed by a message indicating the program finished with exit code 0.

```
1 n = 7
2 arr = [5, 10, 15, 20, 25, 30, 35]
3 key = 25
4
5 low = 0
6 high = n - 1
7 count = 0
8 found = False
9
10 while low <= high:
11     count += 1
12     mid = (low + high) // 2
13
14     if arr[mid] == key:
15         print("Element found")
16         print("Iterations =", count)
17         found = True
18         break
19     elif arr[mid] < key:
20         low = mid + 1
21     else:
22         high = mid - 1
23
24 if not found:
25     print("Element not found")
26     print("Iterations =", count)
```

Element found
Iterations = 3
...Program finished with exit code 0
Press ENTER to exit console.

Question 4: Find the First Occurrence of an Element

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Test Case 1

Input

8

1 2 2 2 3 4 5 6

2

Expected Output

First occurrence at index 1

Test Case 2

Input

8

1 2 2 2 3 4 5 6

7

Expected Output

Element not found

```
main.py
1 n = 6
2 arr = [2, 4, 6, 8, 10, 12]
3 key = 8
4
5 low = 0
6 high = n - 1
7 index = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         index = mid
14         break
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 print("Index =", index)
```

Index = 3

...Program finished with exit code 0
Press ENTER to exit console.

Question 5: Find the Last Occurrence of an Element

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Test Case 1

Input

8
1 2 2 2 3 4 5 6
2

Expected Output

Last occurrence at index 3

Test Case 2

Input

8
1 2 2 2 3 4 5 6
9

Expected Output

Element not found

```
main.py
1 n = 8
2 arr = [1, 2, 2, 2, 3, 4, 5, 6]
3 key = 2
4
5 low = 0
6 high = n - 1
7 ans = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         ans = mid
14         high = mid - 1
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if ans != -1:
21     print("First occurrence at index", ans)
22 else:
23     print("Element not found")
```

First occurrence at index 1

...Program finished with exit code 0
Press ENTER to exit console.

Question 6: Highest and Lowest Temperature Recorded

A weather station records temperatures over a week. Find the **minimum** and **maximum** temperature using the Divide and Conquer approach.

Input

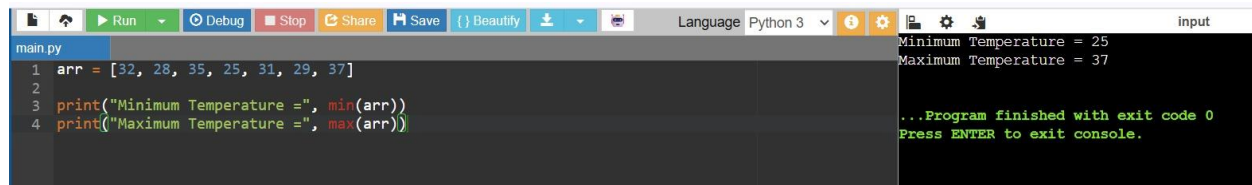
7

32 28 35 25 31 29 37

Expected Output

Minimum Temperature = 25

Maximum Temperature = 37



```
main.py
1 arr = [32, 28, 35, 25, 31, 29, 37]
2
3 print("Minimum Temperature =", min(arr))
4 print("Maximum Temperature =", max(arr))
```

Minimum Temperature = 25
Maximum Temperature = 37
...Program finished with exit code 0
Press ENTER to exit console.

Question 7: Best and Worst Student Marks

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Input

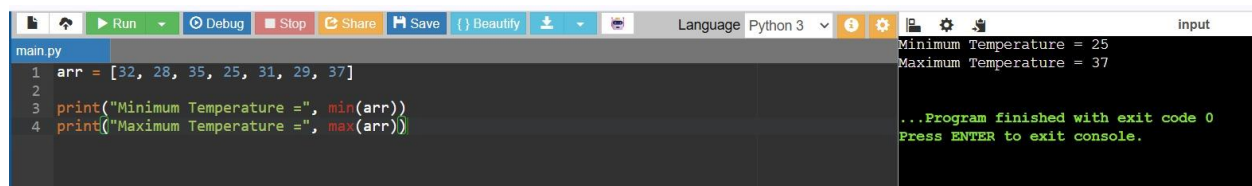
8

78 92 65 88 95 72 81 69

Expected Output

Minimum Mark = 65

Maximum Mark = 95



```
main.py
1 arr = [78, 92, 65, 88, 95, 72, 81, 69]
2
3 print("Minimum Mark =", min(arr))
4 print("Maximum Mark =", max(arr))
```

Minimum Mark = 65
Maximum Mark = 95
...Program finished with exit code 0
Press ENTER to exit console.

Question 8: Stock Price Analysis

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Input

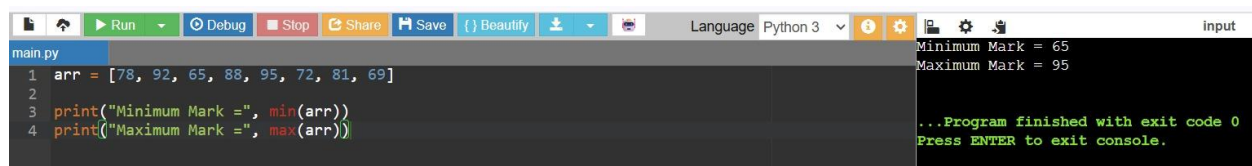
6

450 520 480 610 430 590

Expected Output

Minimum Stock Price = 430

Maximum Stock Price = 610



```
main.py
1 arr = [450, 520, 480, 610, 430, 590]
2
3 print("Minimum Stock Price =", min(arr))
4 print("Maximum Stock Price =", max(arr))
```

Minimum Stock Price = 430
Maximum Stock Price = 610
...Program finished with exit code 0
Press ENTER to exit console.

Question 9: Fastest and Slowest Race Times

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Input

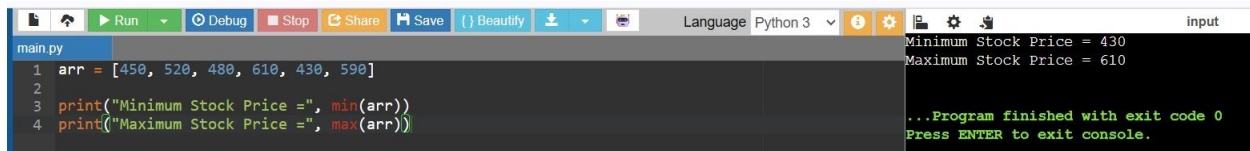
9

14 12 18 11 15 13 17 16 10

Expected Output

Fastest Time = 10

Slowest Time = 18



```
main.py
1 arr = [450, 520, 480, 610, 430, 590]
2
3 print("Minimum Stock Price =", min(arr))
4 print("Maximum Stock Price =", max(arr))
```

```
input
Minimum Stock Price = 430
Maximum Stock Price = 610

...Program finished with exit code 0
Press ENTER to exit console.
```

Question 10: Largest and Smallest Building Heights

An urban planner is analyzing building heights in a city block and wants to determine the tallest and shortest buildings.

Input

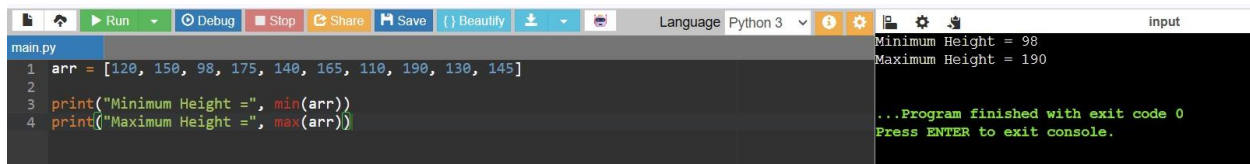
10

120 150 98 175 140 165 110 190 130 145

Expected Output

Minimum Height = 98

Maximum Height = 190



```
main.py
1 arr = [120, 150, 98, 175, 140, 165, 110, 190, 130, 145]
2
3 print("Minimum Height =", min(arr))
4 print("Maximum Height =", max(arr))
```

```
input
Minimum Height = 98
Maximum Height = 190

...Program finished with exit code 0
Press ENTER to exit console.
```